

Quantum-Like Theory of Criminogenic Systems

Crime, Measurement, Conflict, and Institutional Adaptation

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Abstract

This paper develops a quantum-like theory of criminogenic systems. The central claim is not that criminal behavior is literally governed by microscopic quantum physics, but that Hilbert-space probability provides a useful formal language for representing ambiguity, observation, social coupling, institutional classification, and adaptive response. Individuals and organizations are represented by density matrices over lawful, deviant, violent, and rehabilitative behavioral states. Legal institutions are represented by measurement operators; social networks by coupling Hamiltonians; war, finance, inequality, and political fragility by criminogenic operators; and artificial intelligence systems by adversarially robust predictive functions. The paper derives a surveillance-legitimacy threshold theorem showing that surveillance can become counterproductive when adaptive criminal response exceeds deterrent capacity relative to institutional legitimacy. The framework integrates mathematics, economics, finance, quantum science, criminology, political science, geopolitics, international relations, warfare economics, statistics, causal inference, and machine learning into a unified probabilistic model of crime and governance.

The paper ends with “The End”

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1 Introduction

Crime is often modeled as either an individual decision, a social pathology, a rational response to incentives, or an institutional classification. Each view captures part of the phenomenon. A complete theory must simultaneously represent agency, structure, incentives, uncertainty, coercion, networks, measurement, and adaptation.

This paper introduces *Quantum-Like Criminogenic Systems*. The word quantum-like is used in the technical sense of Hilbert-space probability. The framework uses the mathematics of states, projectors, measurement operators, density matrices, and non-commuting observables to model social uncertainty and institutional intervention.

The framework rests on six principles:

1. **Behavioral superposition:** actors can occupy ambiguous mixtures of lawful, deviant, violent, and rehabilitative tendencies.
2. **Institutional measurement:** police, courts, surveillance systems, and welfare agencies classify uncertain actors into legal-administrative categories.
3. **Decoherence:** legal intervention reduces ambiguity by fixing identities such as suspect, offender, convict, acquitted person, or rehabilitated actor.
4. **Network coupling:** criminal and rehabilitative behavior are transmitted through social, financial, and organizational networks.
5. **Political-economic forcing:** poverty, inequality, unemployment, state fragility, conflict, and illicit finance reshape the criminogenic field.
6. **Adaptive response:** criminal systems learn, mutate, and evade in response to surveillance and predictive governance.

The paper is organized as follows. Section 2 defines the behavioral state space. Section 3 formalizes institutional measurement. Section 4 develops criminogenic dynamics. Sections 5–8 integrate economics, finance, geopolitics, war, and networks. Section 9 proves a surveillance-legitimacy threshold theorem. Sections 10–11 develop AI, statistics, and empirical identification. Section 12 discusses normative constraints. Section 13 concludes.

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2 Notation

Table 1: Core notation

Symbol	Meaning
$\mathcal{H}$	Behavioral Hilbert space
$\rho_i(t)$	Density matrix of actor i at time t
$ L\rangle$	Lawful state
$ D\rangle$	Deviant state
$ V\rangle$	Violent state
$ R\rangle$	Rehabilitative state
Π_C	Projector onto criminogenic states
M_y	Institutional measurement operator for outcome y
$H_i(t)$	Criminogenic Hamiltonian of actor i
$L_{ik}(t)$	Shock, intervention, or decoherence operator
$G_t = (V, E_t, W_t)$	Time-varying weighted social graph
$\Lambda_i(t)$	Perceived institutional legitimacy
$S_i(t)$	Surveillance intensity
$K_i(t)$	Rehabilitative capacity
$\mathcal{F}_s(t)$	State fragility index
$\mathcal{A}_t$	Illicit financial adjacency matrix

3 Behavioral State Space

3.1 Hilbert-Space Representation

Let the behavioral basis be

$$\mathcal{B} = \{|L\rangle, |D\rangle, |V\rangle, |R\rangle\},$$

where $|L\rangle$ denotes lawful behavior, $|D\rangle$ deviant behavior, $|V\rangle$ violent behavior, and $|R\rangle$ rehabilitative behavior. The behavioral Hilbert space is

$$\mathcal{H} = \text{span}\{|L\rangle, |D\rangle, |V\rangle, |R\rangle\} \cong \mathbb{C}^4.$$

A pure behavioral state is

$$|\psi_i(t)\rangle = \alpha_i(t)|L\rangle + \beta_i(t)|D\rangle + \gamma_i(t)|V\rangle + \delta_i(t)|R\rangle,$$

with normalization

$$|\alpha_i(t)|^2 + |\beta_i(t)|^2 + |\gamma_i(t)|^2 + |\delta_i(t)|^2 = 1.$$

However, real social actors are usually partially observed, socially mixed, and institutionally noisy. Therefore the primitive object is not a pure vector but a density matrix.

Definition 1 (Behavioral density matrix). *A behavioral state of actor i at time t is a matrix*

$$\rho_i(t) \in \mathbb{C}^{4 \times 4}$$

such that

$$\rho_i(t) \succeq 0, \quad \text{Tr}(\rho_i(t)) = 1.$$

The diagonal entries represent behavioral probabilities, while the off-diagonal entries represent unresolved ambiguity, contextual dependence, and interference between behavioral classifications.

3.2 Crime Probability

Define the projectors

$$\Pi_L = |L\rangle\langle L|, \quad \Pi_D = |D\rangle\langle D|, \quad \Pi_V = |V\rangle\langle V|, \quad \Pi_R = |R\rangle\langle R|.$$

The criminogenic projector is

$$\Pi_C = \Pi_D + \Pi_V.$$

The probability that actor i occupies a criminogenic state at time t is

$$p_i^C(t) = \text{Tr}(\Pi_C \rho_i(t)).$$

Similarly,

$$p_i^L(t) = \text{Tr}(\Pi_L \rho_i(t)), \quad p_i^R(t) = \text{Tr}(\Pi_R \rho_i(t)).$$

The object $p_i^C(t)$ is not a conviction probability. It is a latent behavioral-risk probability. Conviction depends on legal measurement, evidence, procedure, institutional bias, and error.

4 Institutional Measurement

4.1 Legal and Administrative Observables

Institutions do not observe the full behavioral state. They measure proxies: reports, arrests, investigations, financial traces, digital records, witness statements, or algorithmic scores. Let the set of institutional outcomes be

$$\mathcal{Y} = \{0, W, A, C, H\},$$

where 0 means no action, W means watchlist or monitoring, A means arrest or charge, C means conviction, and H means rehabilitation or diversion.

Definition 2 (Institutional measurement). *An institutional measurement is a family of operators*

$$\{M_y : y \in \mathcal{Y}\}$$

satisfying the completeness condition

$$\sum_{y \in \mathcal{Y}} M_y^\dagger M_y = I.$$

The probability of institutional outcome y is

$$\mathbb{P}(Y_i(t) = y \mid \rho_i(t)) = \text{Tr}(M_y^\dagger M_y \rho_i(t)).$$

After observing outcome y , the updated state is

$$\rho_i(t^+) = \frac{M_y \rho_i(t) M_y^\dagger}{\text{Tr}(M_y^\dagger M_y \rho_i(t))}.$$

This is a formal model of institutional classification. It represents the fact that being watched, charged, punished, acquitted, or diverted changes the actor's future social state.

4.2 False Positives and False Negatives

Let C denote the latent criminogenic event and $Y = A$ denote institutional arrest. Then

$$\text{False Positive Rate} = \mathbb{P}(Y = A \mid C = 0),$$

and

$$\text{False Negative Rate} = \mathbb{P}(Y \neq A \mid C = 1).$$

In operator form, if $\Pi_{\bar{C}} = I - \Pi_C$, then

$$\text{FPR} = \frac{\text{Tr}(M_A^\dagger M_A \Pi_{\bar{C}} \rho \Pi_{\bar{C}})}{\text{Tr}(\Pi_{\bar{C}} \rho)},$$

and

$$\text{FNR} = \frac{\text{Tr}((I - M_A^\dagger M_A) \Pi_C \rho \Pi_C)}{\text{Tr}(\Pi_C \rho)}.$$

A legitimate criminogenic prediction system must not only predict; it must be calibrated, audited, contestable, and institutionally constrained.

5 Criminogenic Dynamics

5.1 Criminogenic Hamiltonian

The criminogenic Hamiltonian summarizes structured pressure on actor i :

$$H_i(t) = H_0 + \theta_P P_i(t) \Pi_C + \theta_U U_i(t) \Pi_C + \theta_I I_i(t) \Pi_C + \theta_G G_i(t) \Pi_C - \theta_\Lambda \Lambda_i(t) \Pi_L - \theta_K K_i(t) \Pi_R.$$

Here $P_i(t)$ is poverty exposure, $U_i(t)$ unemployment exposure, $I_i(t)$ inequality exposure, $G_i(t)$ geopolitical instability exposure, $\Lambda_i(t)$ institutional legitimacy, and $K_i(t)$ rehabilitative capacity.

The signs encode the theory:

$$P, U, I, G \quad \text{raise criminogenic pressure,}$$

whereas

$$\Lambda, K \quad \text{stabilize lawful and rehabilitative states.}$$

5.2 Open-System Evolution

Social actors are open systems. They are affected by shocks, surveillance, punishment, trauma, schooling, unemployment, war, migration, and rehabilitation. Therefore their density matrices evolve according to a Lindblad-type equation:

$$\frac{d\rho_i}{dt} = -i[H_i(t), \rho_i(t)] + \sum_k \left(L_{ik}(t) \rho_i(t) L_{ik}^\dagger(t) - \frac{1}{2} \{L_{ik}^\dagger(t) L_{ik}(t), \rho_i(t)\} \right).$$

The commutator term models structured pressure, while the summation term models institutional and environmental shocks.

Proposition 1 (Trace preservation). *If $\rho_i(t)$ evolves according to the Lindblad-type equation above, then*

$$\frac{d}{dt} \text{Tr}(\rho_i(t)) = 0.$$

Proof. Using cyclicity of trace,

$$\text{Tr}([H_i, \rho_i]) = \text{Tr}(H_i \rho_i - \rho_i H_i) = 0.$$

For each k ,

$$\text{Tr}(L_{ik} \rho_i L_{ik}^\dagger) = \text{Tr}(L_{ik}^\dagger L_{ik} \rho_i),$$

and

$$\text{Tr}\left(\frac{1}{2}\{L_{ik}^\dagger L_{ik}, \rho_i\}\right) = \text{Tr}(L_{ik}^\dagger L_{ik} \rho_i).$$

Thus each dissipative term has zero trace. Hence $d \text{Tr}(\rho_i)/dt = 0$. $\square$

6 Economics of Crime

6.1 Expected Utility

Let the expected utility of crime for actor i be

$$U_i^C(t) = p_i^s(t)G_i(t) - p_i^c(t)F_i(t) - R_i(t) + \omega N_i(t) - \chi \Lambda_i(t) - \phi K_i(t),$$

where p_i^s is probability of successful crime, G_i illegal gain, p_i^c probability of capture, F_i punishment cost, R_i reputational cost, N_i criminal network support, Λ_i legitimacy, and K_i rehabilitation capacity.

The transition probability into criminogenic behavior may be represented by the logistic link

$$\mathbb{P}(i \rightarrow C \mid t) = \sigma(U_i^C(t)) = \frac{1}{1 + \exp(-U_i^C(t))}.$$

This connects rational-choice criminology with probabilistic behavioral states.

6.2 Comparative Statics

The marginal effects are

$$\begin{aligned} \frac{\partial U_i^C}{\partial G_i} &= p_i^s(t), & \frac{\partial U_i^C}{\partial F_i} &= -p_i^c(t), \\ \frac{\partial U_i^C}{\partial \Lambda_i} &= -\chi, & \frac{\partial U_i^C}{\partial K_i} &= -\phi. \end{aligned}$$

Thus deterrence is not merely a function of punishment severity. It also depends on capture probability, legitimacy, rehabilitation, and network incentives.

7 Finance and Illicit Network Contagion

Let $\mathcal{A}_t$ be a weighted adjacency matrix of illicit financial flows. The vector $x_t \in \mathbb{R}^n$ records illicit liquidity at nodes such as firms, shell entities, brokers, gangs, officials, or foreign accounts. A linear contagion model is

$$x_{t+1} = \mathcal{A}_t x_t + \varepsilon_t.$$

If $\rho(\mathcal{A}_t)$ is the spectral radius of $\mathcal{A}_t$, then illicit liquidity becomes dynamically persistent when

$$\rho(\mathcal{A}_t) \geq 1.$$

Proposition 2 (Financial containment). *If $\mathcal{A}$ is time-invariant, $\varepsilon_t = 0$, and $\rho(\mathcal{A}) < 1$, then*

$$\lim_{t \rightarrow \infty} x_t = 0.$$

Proof. Since $x_t = \mathcal{A}^t x_0$ and $\rho(\mathcal{A}) < 1$, standard linear systems theory implies $\mathcal{A}^t \rightarrow 0$. Therefore $x_t \rightarrow 0$. $\square$

Financial enforcement therefore targets not only individual crimes but the spectral structure of illicit networks.

8 Political Science, Geopolitics, and International Relations

8.1 State Fragility

Define state fragility as

$$\mathcal{F}_s(t) = a_D D_s(t) + a_E E_s(t) + a_W W_s(t) + a_B B_s(t) - a_L L_s(t) - a_C C_s(t),$$

where D_s is public debt stress, E_s ethnic or social fragmentation, W_s warfare intensity, B_s border permeability, L_s institutional legitimacy, and C_s administrative capacity.

Criminogenic pressure at the macro level is then

$$\mathcal{C}_s(t) = \varphi_0 + \varphi_1 \mathcal{F}_s(t) + \varphi_2 I_s(t) + \varphi_3 U_s(t) + \varphi_4 Q_s(t),$$

where I_s is inequality, U_s unemployment, and Q_s illicit opportunity.

8.2 International Spillovers

Let s and r be states. Cross-border criminogenic transmission can be modeled as

$$\mathcal{C}_s(t+1) = b_{ss} \mathcal{C}_s(t) + \sum_{r \neq s} b_{sr} \mathcal{C}_r(t) + u_s(t).$$

The coefficient b_{sr} measures cross-border transmission through migration, smuggling corridors, financial flows, cybercrime, arms routes, insurgency networks, or sanctions evasion.

8.3 State–Criminal Organization Game

Table 2: Strategic interaction between state and criminal organization

		Criminal organization	
		Escalate	Cooperate
State	Suppress	$(-2, -3)$	$(3, -2)$
State	Negotiate	$(-4, 2)$	$(2, 2)$

The matrix illustrates that repression, bargaining, escalation, and cooperation generate different equilibria depending on enforcement credibility, legitimacy, and the organization's outside options.

9 Warfare Economics and Criminal Amplification

War changes the opportunity structure of crime. It weakens courts, expands black markets, increases arms availability, displaces populations, and creates armed entrepreneurs.

Define the war-crime amplification index:

$$\mathcal{W}_s(t) = \eta_B B_s(t) + \eta_A A_s(t) + \eta_M M_s(t) + \eta_R R_s(t) + \eta_X X_s(t),$$

where B_s is black-market intensity, A_s arms availability, M_s military fragmentation, R_s refugee displacement, and X_s external proxy intervention.

A war shock operator may be written as

$$L_W = \sqrt{\omega_W} (|D\rangle\langle L| + |V\rangle\langle D|),$$

which means that war can move lawful behavior toward deviance and deviance toward violence.

A rehabilitation operator may be written as

$$L_K = \sqrt{\omega_K} (|R\rangle\langle D| + |L\rangle\langle R|),$$

which means that diversion and reintegration can move deviant actors toward rehabilitative and lawful states.

10 Social Entanglement and Network Coupling

Let

$$G_t = (V, E_t, W_t)$$

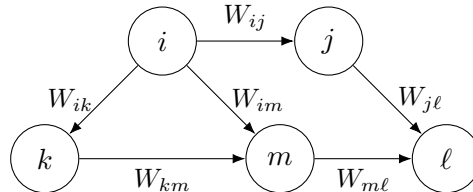
be a weighted social graph. The edge weight $W_{ij}(t)$ measures the strength of interaction between actors i and j . The joint social state is

$$\rho_{1:N}(t) \in \mathcal{D} \left(\bigotimes_{i=1}^N \mathcal{H}_i \right).$$

The interaction Hamiltonian is

$$H_{\text{int}}(t) = \sum_{i < j} W_{ij}(t) \left[\zeta_D \Pi_D^{(i)} \otimes \Pi_D^{(j)} + \zeta_V \Pi_V^{(i)} \otimes \Pi_V^{(j)} - \zeta_R \Pi_R^{(i)} \otimes \Pi_R^{(j)} \right].$$

Positive ζ_D and ζ_V represent deviant and violent peer reinforcement. Positive ζ_R represents rehabilitative peer stabilization.



Network coupling transmits deviant, violent, financial, and rehabilitative influence through weighted social ties.

Figure 1: Network coupling in a criminogenic system

11 Surveillance-Legitimacy Threshold

Surveillance has two opposing effects. It can deter crime by increasing capture probability, but it can also produce adaptation, displacement, evasion, distrust, and mutation.

Assumption 1 (Adaptive response). *Let surveillance intensity be $S > 0$ and institutional legitimacy be $\Lambda > 0$. Adaptive criminal response is*

$$A(S, \Lambda) = \kappa \left(\frac{S}{\Lambda} \right)^\alpha, \quad \kappa > 0, \quad \alpha > 1.$$

Assumption 2 (Deterrence). *Preventive deterrence is linear:*

$$D(S) = \mu S, \quad \mu > 0.$$

Define net adaptation as

$$N(S, \Lambda) = A(S, \Lambda) - D(S).$$

Theorem 1 (Surveillance-legitimacy threshold). *There exists a critical surveillance threshold*

$$S_c = \left(\frac{\mu \Lambda^\alpha}{\kappa} \right)^{1/(\alpha-1)}$$

such that

$$N(S, \Lambda) > 0 \quad \text{if and only if} \quad S > S_c.$$

Proof. Net adaptation is positive exactly when

$$A(S, \Lambda) > D(S).$$

Substituting the functional forms gives

$$\kappa \left(\frac{S}{\Lambda} \right)^\alpha > \mu S.$$

For $S > 0$,

$$\kappa S^\alpha > \mu S \Lambda^\alpha.$$

Dividing by S yields

$$\kappa S^{\alpha-1} > \mu \Lambda^\alpha.$$

Therefore

$$S > \left(\frac{\mu \Lambda^\alpha}{\kappa} \right)^{1/(\alpha-1)} = S_c.$$

Thus surveillance becomes counterproductive precisely beyond the legitimacy-adjusted threshold S_c . □

Corollary 1 (Legitimacy raises the threshold). *If $\alpha > 1$, then*

$$\frac{\partial S_c}{\partial \Lambda} > 0.$$

Hence higher legitimacy permits greater surveillance before adaptive backlash dominates deterrence.

Proof. Since

$$S_c = \left(\frac{\mu}{\kappa} \right)^{1/(\alpha-1)} \Lambda^{\alpha/(\alpha-1)},$$

and $\alpha/(\alpha-1) > 0$, the derivative with respect to Λ is positive. □

12 Artificial Intelligence and Machine Learning

12.1 Predictive Model

Let the predictive system be

$$\hat{p}_i^C(t) = f_\theta(X_i(t), Z_i(t), G_t, M_t),$$

where $X_i(t)$ are individual covariates, $Z_i(t)$ institutional-contact variables, G_t network variables, and M_t macro-political variables.

A prediction loss is

$$\mathcal{L}_{\text{pred}}(\theta) = -\frac{1}{N} \sum_{i=1}^N [Y_i \log \hat{p}_i^C + (1 - Y_i) \log(1 - \hat{p}_i^C)].$$

A responsible predictive system solves

$$\min_{\theta} [\mathcal{L}_{\text{pred}}(\theta) + \lambda_1 \mathcal{L}_{\text{fair}}(\theta) + \lambda_2 \mathcal{L}_{\text{cal}}(\theta) + \lambda_3 \mathcal{L}_{\text{rob}}(\theta)].$$

Here $\mathcal{L}_{\text{fair}}$ penalizes unjust disparity, $\mathcal{L}_{\text{cal}}$ penalizes miscalibration, and $\mathcal{L}_{\text{rob}}$ penalizes adversarial fragility.

12.2 Calibration

For bins $B_1, \dots, B_K$ of predicted risk, calibration requires

$$\mathbb{P}(Y = 1 \mid \hat{p} \in B_k) \approx \frac{1}{|B_k|} \sum_{i \in B_k} \hat{p}_i.$$

A model that is accurate but uncalibrated is dangerous for legal use because its probability scores cannot be interpreted as reliable institutional evidence.

12.3 Adversarial Robustness

Criminal organizations may adapt to predictive systems. The robust learning problem is

$$\min_{\theta} \max_{\delta \in \Delta} \mathcal{L}(f_{\theta}(X + \delta), Y).$$

This expression is used for defensive auditing: to identify whether an institutional model is fragile under strategic behavioral shifts.

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13 Algorithms

Table 3: Algorithm 1: Quantum-like criminogenic risk updating

Step	Operation
Input	Actor covariates $X_i(t)$, institutional variables $Z_i(t)$, graph G_t , macro indicators M_t , prior state $\rho_i(t)$.
1	Construct the criminogenic Hamiltonian $H_i(t)$.
2	Construct shock and intervention operators $L_{ik}(t)$.
3	Evolve $\rho_i(t)$ using the open-system equation.
4	Apply institutional measurement operators M_y .
5	Compute $p_i^C(t) = \text{Tr}(\Pi_C \rho_i(t))$.
6	Update model parameters using observed outcomes.
7	Audit calibration, false positives, false negatives, and group disparity.
Output	Calibrated criminogenic risk distribution and institutional uncertainty report.

Table 4: Algorithm 2: Empirical validation and governance audit

Step	Operation
Input	Panel data on crime, institutions, socioeconomic conditions, conflict, and surveillance.
1	Define treatment, control, and pre-policy periods.
2	Estimate baseline risk model.
3	Test calibration within risk bins.
4	Estimate policy effect using difference-in-differences or event study.
5	Estimate spillovers using graph exposure variables.
6	Conduct fairness audit across legally protected and socioeconomically vulnerable groups.
7	Conduct adversarial stress tests.
8	Produce admissibility report for institutional use.
Output	Validated, contested, and audited policy-relevant model.

14 Statistics and Empirical Identification

14.1 Panel Model

Let Y_{ist} denote crime outcome for actor i in state or region s at time t . A panel specification is

$$Y_{ist} = \alpha_i + \tau_t + \beta S_{st} + \gamma \Lambda_{st} + \delta(S_{st} \times \Lambda_{st}) + X'_{ist}\theta + \varepsilon_{ist}.$$

The interaction term tests whether legitimacy moderates the effect of surveillance.

14.2 Difference-in-Differences

Suppose surveillance expands in treated regions after time T_0 . The difference-in-differences model is

$$Y_{st} = \alpha_s + \tau_t + \beta(\text{Treated}_s \times \text{Post}_t) + X'_{st}\theta + \varepsilon_{st}.$$

The coefficient β estimates the average treatment effect under a parallel trends assumption.

14.3 Event Study

A dynamic event-study design is

$$Y_{st} = \alpha_s + \tau_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - T_s = k\} + X'_{st} \theta + \varepsilon_{st}.$$

Pre-treatment coefficients test pre-trends. Post-treatment coefficients estimate dynamic adaptation or deterrence.

14.4 Graph Exposure

Let

$$E_i(t) = \sum_{j \neq i} W_{ij}(t) Y_j(t)$$

be exposure to criminogenic network behavior. A graph-augmented model is

$$Y_i(t+1) = \alpha + \beta E_i(t) + X_i(t)' \theta + \varepsilon_i(t+1).$$

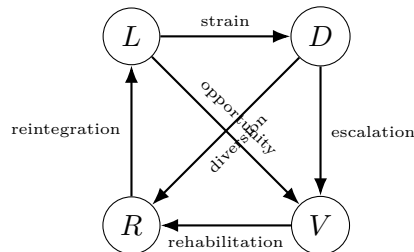
This estimates peer and network spillovers.

Table 5: Empirical variables and observable proxies

Concept	Observable proxy
Poverty exposure	Poverty rate, consumption deprivation, food insecurity
Unemployment	Local unemployment rate, job loss shocks
Inequality	Gini coefficient, income percentile ratio
Surveillance	CCTV density, police stops, digital monitoring exposure
Legitimacy	Trust in police, court confidence, corruption perception
Conflict	Battle deaths, displacement, arms availability
Network exposure	Co-offending graph, neighborhood graph, financial graph
Rehabilitation	Diversion access, schooling, job placement, treatment access
Illicit finance	Suspicious flows, shell-company links, cash-intensive nodes
Crime outcome	Victimization, arrest, charge, conviction, recidivism

15 Vector Graphics

15.1 Behavioral State Geometry



The density matrix $\rho_i(t)$ assigns probabilities and ambiguity to transitions among lawful, deviant, violent, and rehabilitative states.

Figure 2: Quantum-like behavioral state system

15.2 Institutional Measurement Pipeline

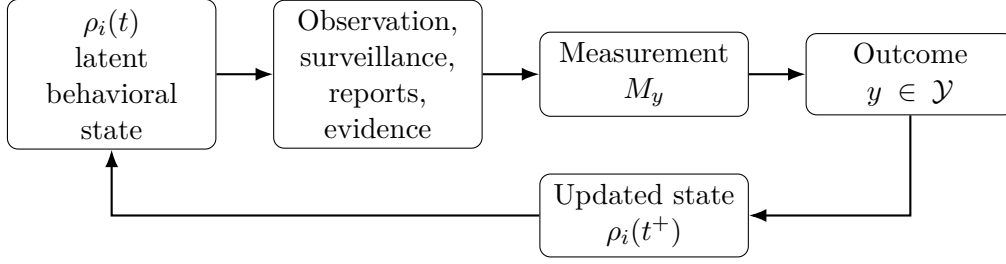


Figure 3: Institutional measurement and state updating

15.3 Political-Economic Criminogenic Field

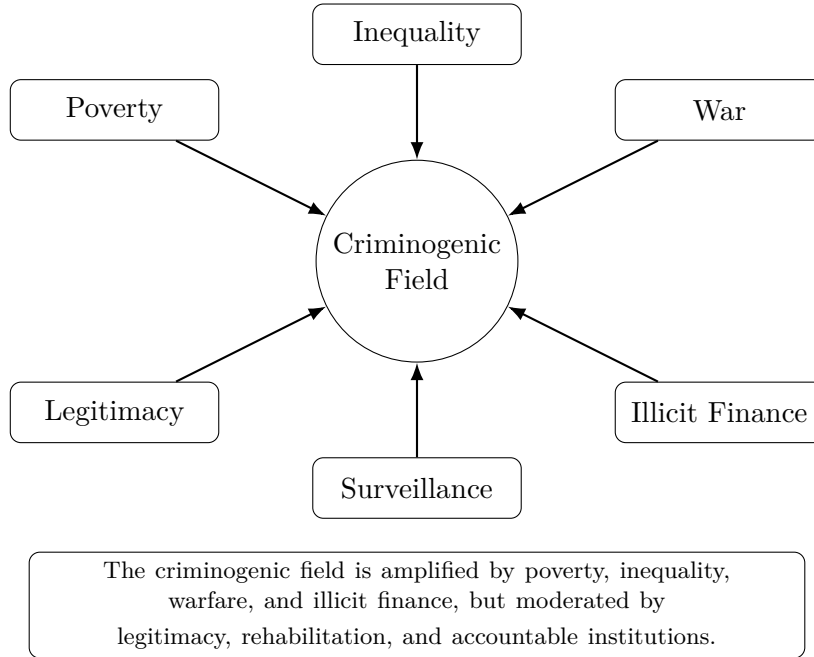


Figure 4: Political-economic drivers of the criminogenic field

16 Policy and Ethics

A criminogenic prediction system is admissible only if it satisfies the following constraints:

$$\text{Admissibility} = \text{Calibration} + \text{Contestability} + \text{Fairness} + \text{Accountability} + \text{Rehabilitation}.$$

The theory rejects purely punitive prediction. The proper institutional objective is

$$\max_{\pi} \mathbb{E} [\text{Public Safety} + \text{Legitimacy} + \text{Rehabilitation} - \text{Rights Violation} - \text{False Punishment}],$$

where π is a policy rule. This objective implies that prediction without procedural justice is not criminological science but administrative risk production.

Table 6: Governance constraints for criminogenic AI systems

Constraint	Institutional meaning
Calibration	Risk scores must correspond to observed frequencies
Contestability	Individuals must be able to challenge classifications
Transparency	Model logic and data provenance must be inspectable
Fairness	Error rates and burdens must be audited across groups
Necessity	Surveillance must be proportionate to legitimate aims
Rehabilitation	Intervention must include non-punitive pathways
Sunset review	Models must expire unless revalidated

17 Discussion

Quantum-like criminology does not replace classical criminology. It generalizes it. Becker-style incentives enter the utility function. Foucaultian surveillance enters the measurement model. Social disorganization enters the network structure. Warfare economics enters the shock operators. International relations enters cross-border spillovers. Machine learning enters predictive inference. Quantum probability supplies the language of uncertain state, measurement, non-commuting observables, and institutional update.

The most important substantive implication is that crime control is not monotonic in surveillance. Surveillance can deter, but it can also mutate criminal behavior, reduce legitimacy, and increase adversarial adaptation. The threshold theorem formalizes this tradeoff.

18 Conclusion

This paper developed a quantum-like theory of criminogenic systems. It represented actors as density matrices over behavioral states, institutions as measurement operators, social networks as coupling Hamiltonians, warfare and finance as criminogenic shocks, and AI systems as adversarially audited predictive models. The framework provides a unified language for crime, uncertainty, legality, political economy, conflict, and institutional adaptation.

The theory’s central policy lesson is that crime control must be legitimacy-constrained. Surveillance without legitimacy can become criminogenic. Prediction without calibration can become arbitrary. Punishment without rehabilitation can preserve the very field it seeks to destroy.

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Glossary

Behavioral density matrix

A positive semidefinite, trace-one matrix representing probabilistic and ambiguous behavioral states.

Calibration

The property that predicted probabilities correspond to observed empirical frequencies.

Criminogenic field

A political-economic and social environment that increases the probability of criminal behavior.

Criminogenic Hamiltonian

An operator representing structured social, economic, political, and geopolitical pressures on behavioral states.

Decoherence

The reduction of behavioral ambiguity through institutional classification, punishment, diversion, or surveillance.

Hilbert-space probability

A probability framework using vectors, operators, projectors, and density matrices.

Institutional legitimacy

The perceived rightful authority of legal and political institutions.

Institutional measurement

The process by which police, courts, surveillance systems, or welfare institutions classify uncertain actors.

Network coupling

The transmission of deviant, violent, financial, or rehabilitative influence through social ties.

Quantum-like criminology

A formal theory using quantum probability as a language for uncertainty, measurement, and social coupling in criminogenic systems.

Surveillance-legitimacy threshold

The level of surveillance beyond which adaptive criminal response exceeds deterrence because legitimacy is insufficient.

War-crime amplification

The process by which conflict expands black markets, arms flows, displacement, and criminal opportunity.

The End